

Trigonometric Substitution

MATH 146 – Calculus II

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April 16, 2025



Overview of Trigonometric Substitution

Integrals involving $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ and $\sqrt{x^2 - a^2}$ can often be solved with a **trigonometric substitution** using the Pythagorean identity:

$$a \cos^2(\theta) + a \sin^2(\theta) = a$$

These substitutions are:

$$\sqrt{a^2 - x^2} \implies x = a \sin(\theta) \implies dx = ???$$

$$\sqrt{a^2 + x^2} \implies x = a \tan(\theta) \implies dx = ???$$

$$\sqrt{x^2 - a^2} \implies x = a \sec(\theta) \implies dx = ???$$

The Bad News

This is a great trick for solving integrals that are very tricky to solve, or even completely unsolvable otherwise.

However, the end result is a *trigonometric integral* in θ .

After you solve the trigonometric integral in θ , you have to convert it back to a solution in terms of x .

Don't forget your inverse trig functions!

Examples: Trig Substitution

Example: Evaluate $\int \frac{1}{\sqrt{4-x^2}} dx$.

Example: Evaluate $\int \frac{1}{4-x^2} dx$. (Is the square root necessary?)

Examples: Trig Substitution

Example: Evaluate $\int \frac{1}{\sqrt{4+x^2}} dx$.

Example: Evaluate $\int \frac{1}{\sqrt{x^2-4}} dx$.

Examples: Trig Substitution

Example: Evaluate $\int \frac{1}{\sqrt{x^2 + 4x - 12}} dx$.

Example: Evaluate $\int \sqrt{x^2 + 9} dx$.